

Practical – 7

Task –1 Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

```
GNU nano 8.7.1 memory_layout.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

void code_function()
{
    printf("This is a code function.\n");
}

int main()
{
    int stack_var = 30;
    static int local_static_var = 40;

    int *heap_var = (int *)malloc(sizeof(int));
    *heap_var = 50;

    printf("Process ID: %d\n\n", getpid());

    printf("Code address      : %p\n", (void *)code_function);
    printf("Global variable   : %p\n", (void *)&global_var);
    printf("Static variable    : %p\n", (void *)&static_var);
    printf("Local static variable : %p\n", (void *)&local_static_var);
    printf("Heap variable      : %p\n", (void *)heap_var);
    printf("Stack variable     : %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");
    printf("Press Ctrl+C to stop.\n");

    while (1)
    {
        sleep(2);
    }

    free(heap_var);

    return 0;
}
```

```
sirichandana@siri:~$ mkdir practical_memory
sirichandana@siri:~$ cd practical_memory
sirichandana@siri:~/practical_memory$ nano memory_layout.c
sirichandana@siri:~/practical_memory$ gcc memory_layout.c -o memory_layout
sirichandana@siri:~/practical_memory$ ./memory_layout
Process ID: 757

Code address      : 0x55c67c7d31c9
Global variable   : 0x55c67c7d6010
Static variable   : 0x55c67c7d6044
Local static variable : 0x55c67c7d6018
Heap variable     : 0x55c6a55f9b10
Stack variable    : 0x7ffc49ae650c

Process is running...
Press Ctrl+C to stop.
sirichandana@siri:~/practical_memory$ █
```

Task-2 Make use of the /proc//maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

void display()
{
    printf("Inside code segment\n");
}

int main()
{
    int stack_var = 30;
    static int local_static = 40;

    int *heap_var = malloc(sizeof(int));
    *heap_var = 50;

    printf("Process ID: %d\n", getpid());

    printf("Code address      : %p\n", (void *)display);
    printf("Global variable    : %p\n", (void *)&global_var);
    printf("Static variable       : %p\n", (void *)&static_var);
    printf("Local static variable : %p\n", (void *)&local_static);
    printf("Heap variable         : %p\n", (void *)heap_var);
    printf("Stack variable        : %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");

    while (1)
    {
        sleep(2);
    }

    free(heap_var);

    return 0;
}
```

```
sirichandana@siri:~/practical_memory$
```

```
sirichandana@siri:~$ cat /proc/1017/maps
```

```
5d9c9ddb3000-5d9c9ddb4000 r--p 00000000 08:30 2739 /home/satwika/memory_analysis
5d9c9ddb4000-5d9c9ddb5000 r-xp 00001000 08:30 2739 /home/satwika/memory_analysis
5d9c9ddb5000-5d9c9ddb6000 r--p 00002000 08:30 2739 /home/satwika/memory_analysis
5d9c9ddb6000-5d9c9ddb7000 r--p 00002000 08:30 2739 /home/satwika/memory_analysis
5d9c9ddb7000-5d9c9ddb8000 rw-p 00003000 08:30 2739 /home/satwika/memory_analysis
5d9cad3d7000-5d9cad3d8000 r--p 00000000 08:00 0 [heap]
7a5d0c6a2000-7a5d0c6a2000 r--p 00000000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7a5d0c6a2000-7a5d0c7c0000 r-xp 00001000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7a5d0c7c0000-7a5d0c800000 r--p 001c0000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7a5d0c800000-7a5d0c812000 r--p 00211000 08:30 13628 /usr/lib/x86_64-linux-gnu/libc.so.6
7a5d0c812000-7a5d0c821000 rw-p 00000000 08:00 0
7a5d0c889000-7a5d0c88c000 r--p 00000000 08:00 0
7a5d0c889000-7a5d0c897000 r--p 00000000 08:00 0
7a5d0c897000-7a5d0c89f000 r-xp 00000000 08:00 0
7a5d0c89f000-7a5d0c8a9000 r--p 00000000 08:00 0
7a5d0c89f000-7a5d0c8a9000 rw-p 00000000 08:00 0
7a5d0c98b000-7a5d0c89c000 r--p 00001000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a5d0c89c000-7a5d0c8d6000 r-xp 00001000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a5d0c8d6000-7a5d0c8d8000 r--p 00032000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a5d0c8d8000-7a5d0c8da000 r--p 00032000 08:30 13625 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a5d0c8da000-7a5d0c8dc000 rw-p 00000000 08:00 0
7ffde3997000-7ffde39b8000 rw-p 00000000 08:00 0 [stack]
```

```
sirichandana@siri:~$
```